BRAIN TUMOR DETECTION AND LOCALIZATION USING YOLO

A MINI PROJECT-II REPORT

Submitted by

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of

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(AUTONOMOUS)

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KONGUNADU COLLEGE OF ENGINEERING AND TECHNOLOGY

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"To become an Internationally Renowned Institution in Technical Education, Research and Development by Transforming the Students into Competent Professionals with Leadership Skills and Ethical Values."

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- Providing the Best Resources and Infrastructure.
- Creating Learner centric Environment and continuous –Learning.
- Promoting Effective Links with Intellectuals and Industries.
- Enriching Employability and Entrepreneurial Skills.
- Adapting to Changes for Sustainable Development.

COMPUTER SCIENCE AND ENGINEERING

VISION

To produce competent software professionals, academicians, researchers and entrepreneurs with moral values through quality education in the field of Computer Science and Engineering.

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PROGRAM OUTCOMES (PO's)

1. **Engineering knowledge**: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **Problem analysis**: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. **Design/development of solutions**: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. **Conduct investigations of complex problems**: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. **Modern tool usage**: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
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7. **Environment and sustainability**: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. **Ethics**: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. **Individual and team work**: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. **Communication**: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. **Project management and finance**: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team,to manage projects and in multidisciplinary environments.
12. **Life Long Learning:** Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM EDUCATIONAL OBJECTIVES (PEO’s)

PEO I: Graduates shall be professionals with expertise in the fields of Software Engineering, Networking, Data Mining and Cloud computing and shall undertake Software Development, Teaching and Research.

PEO II: Graduates will analyzes problems, design solutions and develop programs with sound Domain Knowledge.

PEO III: Graduates shall have professional ethics, team spirit, life-long learning, good oral and written communication skills and adopt corporate culture, core values and leadership skills.

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PSO2: Competency: Students shall qualify at the State, National and International level competitive examination for employment, higher studies and research.

KONGUNADU COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

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BONAFIDE CERTIFICATE

Certified that this Mini Project-II report **"AI-DRIVEN BRAIN TUMOR DETECTION AND LOCALIZATION USING YOLO"** is the Bonafide work of **DHANARAJ R (621322104019), GOWTHAM S (621322104029), NITHISH S J (621322104061),** who carried out the Mini Project-II work under my supervision.

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Submitted for the Mini Project-II Viva-Voce examination held on_____________

INTERNAL EXAMINER **EXTERNAL EXAMINER**

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ABSTRACT

Brain tumors remain one of the most severe health challenges, necessitating early and accurate diagnosis to enhance treatment effectiveness and patient survival rates. This project introduces an advanced AI-powered system for brain tumor detection and localization using the state-of-the-art **You Only Look Once version 8 (YOLOv8)** framework. YOLOv8 is a real-time object detection algorithm renowned for its speed and precision, making it highly suitable for analyzing complex medical imaging data, such as MRI and CT scans. The model was trained using YOLOv8 on a well-annotated and diverse dataset of brain scan images, enabling it to effectively learn and recognize tumor patterns with high accuracy. Through the use of data augmentation techniques and rigorous evaluation metrics such as **mean Average Precision (mAP)** and **Intersection over Union (IoU)**, the model was optimized to reduce false positives and negatives while improving detection precision and reliability. This deep learning-based approach enables accurate and efficient identification of tumor regions, significantly reducing diagnostic delays. By leveraging YOLOv8's capabilities, the system contributes to enhancing diagnostic support in medical imaging and demonstrates the potential of AI in assisting clinical decision-making for improved patient care.

TABLE OF CONTENTS

LIST OF FIGURES

LIST OF ABBREVIATIONS

AI	-	**A**rtificial **I**ntelligence
API	-	**A**pplication **P**rogramming **I**nterface
DOM	-	**D**ocument **O**bject **M**odel
F1	-	**F1** Score
FTP	-	**F**ile **T**ransfer **P**rotocol
GIT	-	**G**lobal **I**nformation **T**racker
GUI	-	**G**raphical **U**ser **I**nterface
HTTP	-	**H**yper **T**ext **T**ransfer **P**rotocol
IDE	-	**I**ntegrated **D**evelopment **E**nvironment
IP	-	**I**nternet **P**rotocol
JSON	-	**J**avaScript **O**bject **N**otation
ML	-	**M**achine **L**earning
OS	-	**O**perating **S**ystem
RAM	-	**R**andom **A**ccess **M**emory
URL	-	**U**niform **R**esource **L**ocator
Wi-Fi	-	**W**ireless **F**idelity
YOLO	-	**Y**ou **O**nly **L**ook **O**nce

CHAPTER 1

INTRODUCTION

12.1 OVERVIEW

Brain tumors pose a significant health challenge, requiring early and accurate detection for effective treatment. Traditional diagnostic methods often suffer from slow detection times and limited precision in identifying tumor boundaries. To address these limitations, this project leverages artificial intelligence (AI) and deep learning techniques, specifically the YOLOv8 framework, to develop an efficient system for real-time brain tumor detection and localization in MRI and CT scans.

The proposed system utilizes YOLOv8's advanced object detection capabilities, enabling rapid and precise identification of tumor regions in medical images. The model is trained on a diverse dataset to ensure high accuracy, minimal false positives, and reliable tumor segmentation. By integrating deep learning with advanced feature extraction, this approach enhances diagnostic efficiency, significantly reducing the time required for tumor identification and improving overall detection reliability.

This AI-powered system provides a cost-effective and scalable solution for healthcare professionals, assisting doctors in making faster and more informed clinical decisions. With its ability to accurately localize tumors in real-time, this approach not only improves diagnostic outcomes but also contributes to better patient management and treatment planning.

12.2 PROBLEM STATEMENT

In today's fast-paced world, the early and accurate diagnosis of life-threatening diseases remains a significant challenge. Brain tumors, in particular, require timely detection for effective treatment and improved patient outcomes.

While advancements in medical imaging have enhanced diagnosis, existing methods still rely heavily on manual interpretation by radiologists, which can be time-consuming and prone to human error. Additionally, traditional tumor detection systems lack real-time processing capabilities, delaying critical decision-making in clinical settings.

Recent developments in artificial intelligence (AI) and deep learning, particularly in real-time object detection, have shown promise in enhancing medical diagnostics. The You Only Look Once (YOLO) framework has demonstrated high accuracy and efficiency in identifying objects in images, making it a viable solution for medical imaging applications. However, most current implementations of AI in medical imaging focus on classification rather than precise localization of tumors, limiting their effectiveness in guiding treatment plans.

The biggest challenge lies in the lack of real-time, automated tumor detection systems capable of accurately identifying brain tumors in MRI and CT scans. Misdiagnosis or delayed detection can lead to ineffective treatment, negatively impacting patient survival rates. While deep learning models have been explored for medical imaging, an efficient, real-time system specifically designed for precise tumor segmentation and localization remains underdeveloped.

The proposed AI-driven Brain Tumor Detection System aims to bridge this gap by leveraging YOLOv8 for real-time tumor identification and localization in MRI and CT scans. By training the model on a diverse dataset, the system ensures high accuracy, minimal false positives, and reliable segmentation. This technology will enable faster and more precise diagnosis, assisting healthcare professionals in making informed decisions and ultimately improving patient care.

12.3 MULTI-ROLE SYSTEM

1. **Tumor Detection Role:**

   By utilizing YOLOv8, the system analyzes MRI and CT scan images to detect and localize brain tumors with high precision. This real-time detection process enables quick identification of tumor regions, assisting doctors in making timely and accurate diagnoses.

2. **Diagnosis Assistance Role:**

   Based on the detected tumor, the system provides insights into its location, size, and possible severity. By reducing manual analysis time, the system helps healthcare professionals streamline the diagnostic process and improve treatment planning.

3. **User Interaction Role:**

   The system ensures a user-friendly interface, allowing radiologists and medical professionals to easily upload scans and receive instant results. The intuitive design enhances accessibility, ensuring that users can efficiently navigate and interpret the results.

4. **Data Handling and Learning Role:**

   The model continuously improves through deep learning and dataset expansion, refining its accuracy in tumor detection. All medical data is processed securely, ensuring patient privacy and compliance with healthcare regulations while enhancing detection performance over time.

CHAPTER 2

LITERATURE SURVEY

[1] Title: Minimizing Certificate Verification Time in Blockchain Technology. Ankit K C, Rhishav Pandey, Deepesh Bhandari, Birendra Khadka, Rojalina Priyadarshini (2024).

The project "Minimizing Certificate Verification Time in Blockchain Technology" intends to enhance the efficiency of verifying a digital certificate on a blockchain platform. By implementing a faster verification algorithm onto the blockchain, it will require less computational resources compared to traditional verification processes, thereby producing quicker transaction confirmations and closer to real-time validation. The project extracts speed benefits by factorizing redundant computations and operational tasks too, while using parallel processing combined with caching techniques. The system also employs more efficient consensus mechanisms that provide quicker values of transaction confirmations without sacrificing security. Lightweight smart contracts are part of the solution for operational stability of the verification of certificate processes. All of which results scalable improvements in verification time capacity and optimized resource utilization, while offering energy efficiency designed to withstand an increase in verification events credibly.

Techniques: Optimized Algorithm, parallel processing, and caching techniques to optimize verification time. Efficient consensus mechanism, and lightweight smart contracts to ensure quicker and secure validations of certificates.

Merits: Time saved on verification. Reducing latency, and optimized resource utilization for scalability, as well as improved user experience of generally smoother user interaction with quicker certificate validations.

Demerits: Complex implementation, and possibly more challenges with security when combining with public blockchains. Security risks.

[2] Title: A YOLOv5 Deep Neural Network Model to Detect Brain Tumor in Portable Electromagnetic Imaging System (2021) Amran Hossain, Mohammad Tariqul Islam, Mohammad Shahidul Islam, Muhammad E.H. Chowdhury.

The project "A YOLOv5 Deep Neural Network Model to Detect Brain Tumor in Portable Electromagnetic Imaging System" presents a novel brain tumor detection system using a deep learning model trained on electromagnetic signal data. Built on the YOLOv5 architecture, the system processes signal patterns to detect tumor regions, offering a real-time, cost-effective alternative to traditional imaging methods. This approach aims to deliver accessible early diagnosis tools for brain tumors, particularly in remote or under-resourced regions where MRI and CT scanning facilities may be unavailable or expensive. By integrating the model into portable medical devices, the solution addresses accessibility gaps while maintaining efficiency in tumor classification.

Techniques: YOLOv5-based deep neural network for signal pattern recognition, electromagnetic imaging for data input, and real-time classification of tumor regions.

Merits: Portable and low-cost solution for early tumor detection. Especially valuable in remote or resource-constrained environments. Provides real-time results with minimal infrastructure needs.

Demerits: Limited testing on large and diverse datasets, requiring broader validation. Relies on electromagnetic imaging, which differs from conventional clinical practices and may face challenges in standard clinical adoption.

[3] Title: A Computer-Aided Diagnosis of Brain Tumors Using a Fine-Tuned YOLO-based Model with Transfer Learning (2022), Grace Shalini J, Jose Daisy Premila M, Rajan, Sharan Basheer, Manikandan.

The project "A Computer-Aided Diagnosis of Brain Tumors Using a Fine-Tuned YOLO-based Model with Transfer Learning" introduces a deep learning solution tailored for efficient and accurate brain tumor detection using MRI scans. By fine-tuning a YOLO (You Only Look Once) object detection model and applying transfer learning techniques, the system enhances both speed and diagnostic precision. Designed for real-time implementation, the model provides fast tumor localization without requiring extensive computational resources. The CAD system is optimized for use on mid-range hardware, making it an accessible and scalable diagnostic aid, especially in settings with limited access to advanced computing infrastructure.

Techniques: Fine-tuned YOLO-based real-time object detection, transfer learning for improved performance on medical datasets, optimized for MRI image input.

Merits: High computational efficiency with reduced dependency on high-end GPUs. Enables real-time and accurate tumor detection, supporting broader accessibility in clinical environments.

Demerits: Limited depth in feature extraction compared to deeper convolutional neural network architectures. May not capture subtle or complex tumor characteristics, which could affect precision in more intricate cases.

[4] Title: Diagnosis of Brain Tumor Using Lightweight Deep Learning Model with Fine-Tuning Approach (2022) Kumar N., Gupta M.

The project "Diagnosis of Brain Tumor Using Lightweight Deep Learning Model with Fine-Tuning Approach" focuses on developing an efficient and adaptable brain tumor detection system optimized for deployment in resource-constrained environments. Utilizing a lightweight deep learning architecture, the model aims to deliver high accuracy while minimizing computational overhead, making it suitable for mobile, embedded, or low-power devices. Through fine-tuning, the system adapts effectively to variations across different MRI datasets, ensuring improved generalization and detection reliability even with limited data availability.

Techniques: Lightweight deep learning models integrated with fine-tuning methods to optimize tumor detection and classification. Designed for minimal computational cost and high adaptability.

Merits: High detection accuracy despite reduced model complexity. Optimized for use in mobile and embedded systems, enabling broader access to diagnostic tools in remote or low-resource settings.

Demerits: Limited feature extraction capability compared to deeper neural networks. Model performance may vary with dataset diversity and requires further testing on large and heterogeneous medical datasets.

[5] Title: RCS-YOLO: A Fast and High-Accuracy Object Detector for Brain Tumor Detection (2023) Ming Kang, Chee-Ming Ting, Fung Fung Ting, Raphaël C.-W. Phan.

The study "RCS-YOLO: A Fast and High-Accuracy Object Detector for Brain Tumor Detection" introduces a cutting-edge YOLO-based deep learning model, RCS-YOLO, designed for rapid and precise brain tumor detection from MRI images. The model leverages a novel Reparameterized Convolution and Channel Shuffle (RCS) technique to significantly improve both the efficiency and accuracy of tumor localization while maintaining high computational speed. Trained on the Br35H brain tumor dataset, RCS-YOLO achieves real-time performance with an impressive inference speed of 114.8 FPS, positioning it as a strong candidate for clinical environments requiring fast diagnostic tools.

Techniques: RCS-OSA (One-Shot Aggregation) module for enhanced semantic information extraction, improving tumor localization. Trained and evaluated on the Br35H brain tumor dataset.

Merits: Achieves high-speed inference (114.8 FPS) suitable for real-time clinical applications. Optimized memory usage reduces computational load, making it an efficient solution for large-scale deployment.

Demerits: Limited generalization across different MRI datasets due to its evaluation on the Br35H dataset. The architecture is more complex than traditional YOLO models, which can increase implementation difficulty.

[6] Title: Automated Brain Tumor Segmentation and Classification in MRI Using YOLO-Based Deep Learning (2024) Maram Fahaad Almufareh, Muhammad Imran.

The study "Automated Brain Tumor Segmentation and Classification in MRI Using YOLO-Based Deep Learning" introduces an advanced YOLO-based model designed for the automatic segmentation and classification of brain tumors in MRI images. This approach aims to enhance both the speed and accuracy of tumor detection, minimizing the manual effort required from radiologists. Leveraging the real-time object detection capabilities of YOLO, the model offers efficient tumor localization and categorization, providing a fast and accurate tool for medical professionals in clinical settings.

Techniques: YOLO-based real-time segmentation and classification for brain tumor detection. Utilizes the YOLO architecture to both detect and segment tumors from MRI scans.

Merits: Fast and efficient detection with real-time processing, significantly reducing the time required for tumor diagnosis and minimizing radiologists' workload.

Demerits: Limited generalization to diverse MRI datasets, which requires additional validation across multi-institutional data. The model may struggle with the segmentation of irregularly shaped or very small tumors, affecting its overall accuracy in certain cases.

CHAPTER 3

SYSTEM ANALYSIS

3.1 EXISTING SYSTEM

Traditional Brain Tumor Detection Methods rely on manual examination of MRI and CT scans by radiologists. These methods require expertise and are time-consuming, often leading to delays in diagnosis and potential human error. Radiologists analyze features such as tumor shape, size, and contrast variations to determine abnormalities. While manual diagnosis remains the gold standard, it lacks real-time processing capabilities and may vary based on the experience of the medical professional.

Deep Learning-Based Tumor Detection Systems use Convolutional Neural Networks (CNNs) and object detection models to automate tumor identification. These systems extract important visual features from MRI and CT scans to detect and classify tumors with high accuracy. Many research studies have explored CNN-based models for medical image analysis, demonstrating promising results in brain tumor detection. However, most models focus on classification rather than precise tumor localization, limiting their effectiveness in clinical decision-making.

Medical Image Segmentation Systems apply techniques like U-Net, Mask R-CNN, and other deep learning architectures to segment tumor regions. These models aim to provide more accurate tumor boundary identification, assisting in treatment planning. While they enhance detection precision, they often require extensive computational resources and fine-tuning to achieve reliable results. Additionally, many segmentation models lack real-time processing, making them impractical for urgent medical scenarios.

3.1.1 DISADVANTAGES

- Lack of Real-Time Tumor Detection
- Limited Tumor Localization Accuracy
- High False Positives and False Negatives
- Computational Complexity
- Latency in Processing Medical Images
- Lack of Integration with Clinical Workflows
- Dependence on Limited Datasets

3.2 PROPOSED SYSTEM

The AI-Driven Brain Tumor Detection System is an advanced tool developed to support early diagnosis and improve treatment planning for brain tumor patients. It leverages real-time analysis of MRI and CT scans to detect tumors quickly and accurately. At the core of the system is YOLOv8, a cutting-edge deep learning model known for its speed and precision in object detection tasks. This powerful framework allows the system to accurately identify and localize brain tumors, helping healthcare professionals make well-informed clinical decisions. By providing high-speed processing and precise tumor boundary detection, the system significantly enhances diagnostic accuracy and supports timely medical intervention.

The system comprises two main modules: Tumor Detection and Localization. The Tumor Detection module processes MRI and CT scans, utilizing deep learning-based feature extraction to identify abnormal growths. The Localization module precisely marks tumor regions, enabling radiologists to assess the tumor's size, shape, and severity for better treatment planning. This dual-module approach ensures efficient and reliable tumor identification, reducing the chances of misdiagnosis.

In addition to tumor detection, the system offers seamless integration with medical workflows, allowing doctors to upload scans and receive instant diagnostic insights. This automated detection process significantly reduces diagnostic delays, enabling faster medical interventions. The system’s real-time analysis capability sets it apart from traditional methods, which rely on manual assessment and classification. By improving speed and accuracy, the system enhances clinical decision-making and patient outcomes.

To ensure data security and compliance with medical privacy standards, the system follows strict encryption and data protection protocols. Patient scan data is processed securely, ensuring that all medical information remains confidential. Overall, the AI-Driven Brain Tumor Detection System provides a comprehensive, real-time, and scalable solution for medical imaging, revolutionizing the way brain tumors are detected and diagnosed.

3.2.1 ADVANTAGES

- Real-Time Tumor Detection
- High Precision and Accuracy
- Automated and Efficient Diagnosis
- Scalability and Adaptability
- Data Security and Privacy

CHAPTER 4

REQUIREMENTS SYSTEM

4.1 HARDWARE REQUIREMENTS

Processor	-	Intel i3 / AMD Ryzen 3 or above
RAM	-	4 GB of RAM
GPU	-	NVIDIA GTX 1050 / 1650
Operating System	-	Windows / macOS / Ubuntu

4.2 SOFTWARE REQUIREMENTS

Programming Language	-	Python
Deep Learning Framework	-	Pytorch
Web Browser	-	Google Chrome or Microsoft Edge
Object Detection Algorithm	-	YOLOv8
Image processing	-	OpenCV
Web Framework	-	Flask
Database	-	MongoDB

CHAPTER 5

SYSTEM DESIGN

5.1 ARCHITECTURE DIAGRAM

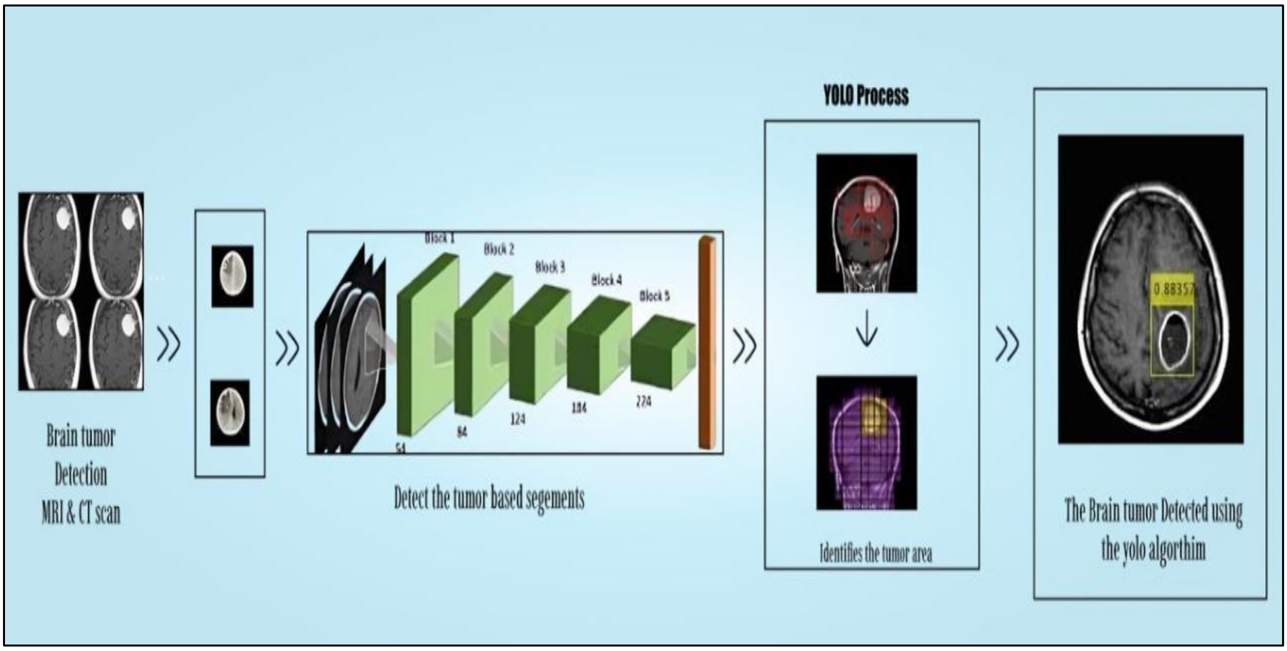

Fig.1 Architecture Diagram

The image illustrates the workflow of an AI-powered Brain Tumor Detection System using MRI and CT scans. The process begins with the input of brain scan images, which are preprocessed to enhance image quality and extract key features. Next, a segmentation process identifies tumor-affected regions for further analysis. The YOLO (You Only Look Once) object detection framework is then applied to localize and classify the tumor with high accuracy. The system processes the segmented images, detects potential tumor areas, and assigns a confidence score to each detection. Finally, the detected tumor is highlighted in the scan, aiding radiologists and doctors in diagnosis. This AI-driven approach enhances speed, accuracy, and efficiency in tumor detection, reducing manual effort and enabling early diagnosis. The system ensures real-time tumor localization, making it a scalable and effective solution for medical imaging analysis in healthcare.

5.2 DATA FLOW DIAGRAM

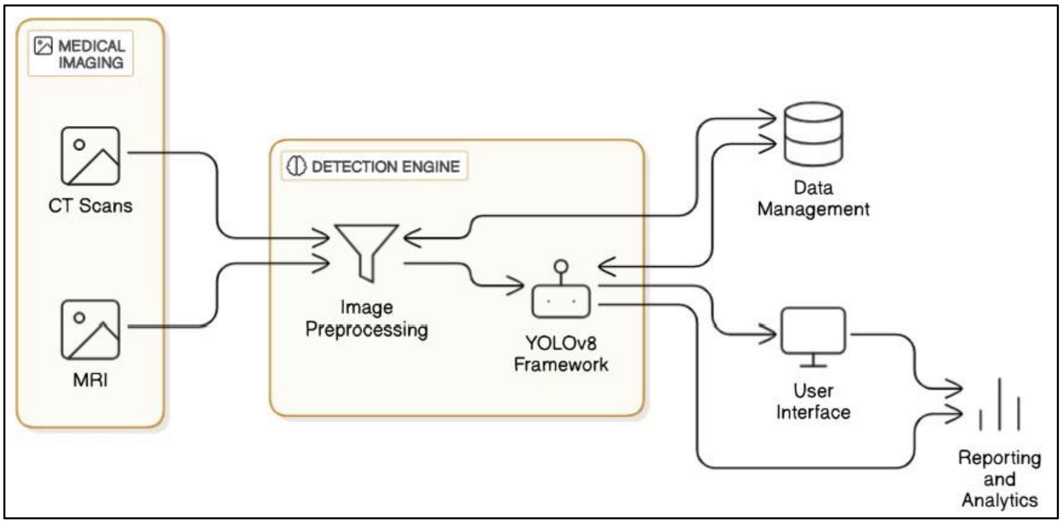

Fig.2 Data flow Diagram

The system architecture for brain tumor detection integrates medical imaging, deep learning, and data management to enhance diagnostic accuracy. It begins by acquiring CT scans and MRI images, which serve as input data for analysis. These images are then processed in the detection engine, where image preprocessing techniques are applied to enhance clarity and remove noise. The preprocessed images are fed into the YOLOv8 framework, a deep learning-based object detection model, which identifies potential brain tumors with high precision. Once the tumor detection process is complete, the results are directed to two key modules: data management and the user interface. The data management module ensures efficient storage, retrieval, and organization of diagnosed cases, facilitating future reference and comparative analysis. Meanwhile, the user interface provides an intuitive platform for medical professionals to visualize and interact with the detection results. Additionally, the system includes a reporting and analytics module, which generates insights, trends, and performance metrics to aid in clinical decision-making and medical research.

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5.3 GRAPHICAL REPRESENTATION

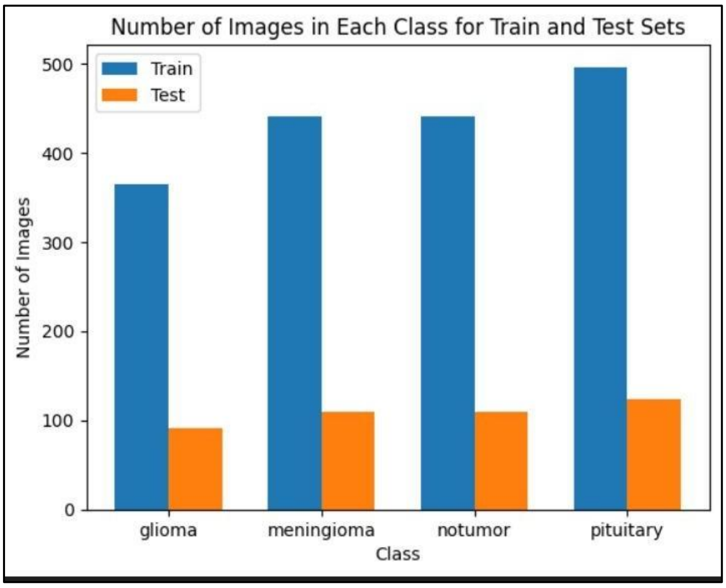

Fig.3 Dataset statistics

The image is a bar chart titled "Number of Images in Each Class for Train and Test Sets", which provides a detailed visualization of the distribution of images across four classes of brain tumors—glioma, meningioma, no tumor and pituitary —in both the training and test sets of a dataset. The x-axis categorizes the tumor classes, while the y-axis quantifies the number of images per class, ranging from 0 to approximately 500. The chart employs two distinct colors to differentiate between the subsets:

- Blue bars represent the training set, which contains the data used for model training.
- Orange bars represent the test set, which serves as an independent dataset for evaluating the model's performance.

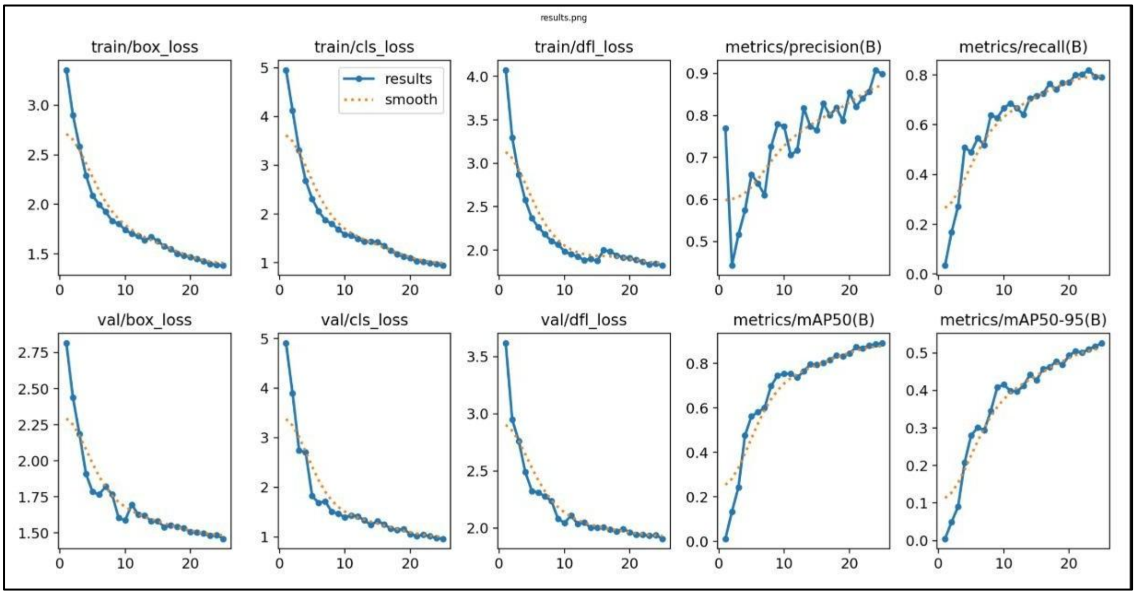

Fig.4 train/val_accuracy

The image contains eight subplots in a 2x4 grid, showcasing training and validation metrics for an object detection model. The top row displays training losses: `train/box_loss`(bounding box regression), `train/cls loss` (classification), and `train/dfl loss`(distribution focal loss), all of which decrease over epochs, indicating effective learning. The fourth plot, `metrics/precision(B)`, shows increasing precision, reflecting improved accuracy in predictions. The bottom row presents validation metrics: `val/box_loss, `val/cls_loss and `val/dfl_loss, which also decrease, confirming strong generalization to unseen data. The final plot, `metrics/mAP50-95(B, shows rising mean Average Precision, highlighting enhanced detection performance across IoU thresholds. Collectively, these plots demonstrate the model's convergence, robustness, and improving accuracy during training and validation.

CHAPTER 6

SYSTEM IMPLEMENTATION

6.1 MODULES

The proposed system for AI-driven brain tumor detection can be divided into six key modules.

- Data Collection
- Data processing
- Model Training
- Tumor Detection and Localization
- Diagnosis Assistance Module
- User Interface Module

6.1.1 Data Collection :

This module is responsible for gathering MRI and CT scan images for analysis. The collected medical images undergo preprocessing to enhance their quality and remove noise. OpenCV is used for image enhancement techniques, while annotation tools help label tumor regions. A well- structured dataset is created to ensure the model can accurately distinguish between normal and tumor-affected scans.

6.1.2 Data Processing :

In this module, medical image preprocessing techniques are applied to ensure that the data is clean and normalized. Contrast enhancement, edge detection, and noise reduction methods improve the clarity of MRI and CT scans. Additionally, dataset augmentation techniques such as rotation, flipping, and intensity scaling help improve the model’s ability to generalize across various medical images.

6.1.3 Model Training :

The YOLOv8 model is trained on a labeled dataset containing tumor and non-tumor MRI/CT scans. Using frameworks like TensorFlow and PyTorch, the model learns to detect tumors by identifying patterns, shapes, and intensities. The training process involves backpropagation, loss function optimization, and fine-tuning hyperparameters to achieve the best detection accuracy. Transfer learning techniques may also be used to improve performance on medical images.

6.1.4 Tumor Detection and Localization :

This module is responsible for real-time tumor detection using the trained YOLOv8 model. When an MRI or CT scan is input, the system identifies and highlights tumor regions with bounding boxes and segmentation masks. The detected tumor's size, shape, and position are extracted, allowing for accurate localization, which is critical for treatment planning.

6.1.5 Diagnosis Assistance :

Once the tumor is detected, this module provides insights into the severity and classification of the tumor. It can assist radiologists by generating automated diagnostic reports and offering second-opinion support. The system integrates with hospital databases or cloud storage, enabling doctors to access scan results efficiently.

6.1.6 User Interface Module :

Designed to be intuitive and interactive, this module allows medical professionals to upload MRI and CT scans, view detection results, and generate diagnostic reports. Built using Django for the backend and React for the frontend, the interface ensures smooth usability and integration with medical imaging systems. The user-friendly dashboard presents tumor detection results in a clear and interpretable format, helping doctors make
quick decisions.

CHAPTER 7

TESTING

The project testing phase focused on ensuring optimal efficiency, accuracy, reliability, and user-focused feedback. This chapter outlines the various stages and methodologies used in testing.

7.1 SYSTEM TESTING

The proposed system implemented in the AI-driven brain tumor detection system can be divided into five categories:

- Unit Testing
- Integration Testing
- Functional Testing
- Acceptance Testing
- Performance Testing

7.1.1 UNIT TESTING

Unit testing focuses on verifying that individual components of the system function correctly in isolation. For example, the MRI scan preprocessing module and tumor detection module are tested separately. Test cases include:

Tumor Detection on MRI Scan

Description: Detects the presence of a tumor in an MRI scan and marks its location.

Pre-condition: The system receives an MRI scan as input.

Expected Result: The system correctly identifies the tumor and highlights its location.

Pass Criteria: The detected tumor region matches the ground truth with high accuracy.

7.1.2 INTEGRATING TESTING

Integration testing ensures that different modules work together seamlessly. In this project, the image preprocessing, YOLOv8 detection, and report generation modules must function as a cohesive unit. Test cases include:

Detection and Report Generation

Description: Ensures the tumor detection module correctly integrates with the diagnosis report generator.

Pre-condition: The system processes an MRI scan.

Expected Result: The system should detect the tumor and generate a diagnostic report.

Pass Criteria: The system accurately detects tumors and generates a report with relevant details.

7.1.3 FUNCTIONAL TESTING

Functional testing verifies that the system’s features meet the defined requirements. It ensures the detection of tumor size, shape, and localization, meeting the standards for clinical applications. Test cases include:

Tumor Size Estimation

Description: Verifies the system's ability to measure and estimate tumor dimensions.

Pre-condition: The system processes an MRI scan with a visible tumor.

Expected Result: The system should provide accurate tumor size measurements.

Pass Criteria: Tumor size estimation aligns with radiologist findings.

7.1.4 ACCEPTANCE TESTING

Acceptance testing ensures that the system meets user expectations and is ready for clinical use. It verifies tumor detection accuracy and system usability for healthcare professionals. Test cases include:

Medical Data Privacy Compliance

Description: Ensures that the system complies with healthcare data security standards (HIPAA, GDPR).

Pre-condition: The system processes and stores MRI scan data.

Expected Result: Data is securely stored and accessed only by authorized personnel.

Pass Criteria: The system meets privacy regulations and maintains secure patient data storage.

7.1.5 PERFORMANCE TESTING

Performance testing evaluates the speed, response time, and stability of the system under different conditions, ensuring real-time processing efficiency. Test cases include:

System Response Time

Description: Measures how quickly the system detects and localizes tumors.

Pre-condition: The system receives an MRI scan as input.

Expected Result: The system should provide detection results within 5 seconds.

Pass Criteria: Tumor detection and localization are completed within the expected timeframe.

CHAPTER 8

CONCLUSION AND FUTURE ENHANCEMENTS

8.1 CONCLUSION

The AI-driven Brain Tumor Detection System integrates YOLOv8, a powerful deep learning model, to enable accurate and real-time identification of brain tumors in MRI and CT scans. This enhances the speed and precision of diagnosis, helping doctors make quicker and more informed decisions. By automating the detection and localization process, the system reduces the need for manual analysis and lowers the risk of human error. The system was carefully tested at multiple stages, including unit, integration, system, and functional levels, to ensure it performs reliably and meets the standards required for use in medical settings. The results show that the system performs with high accuracy, improving early detection and reducing the workload for medical professionals. Overall, it provides a scalable, efficient, and trustworthy solution for brain tumor diagnosis in modern healthcare.

8.2 FUTURE ENHANCEMENTS

Although the proposed Brain Tumor Detection System demonstrates high accuracy and efficiency in tumor identification, there are several areas for future enhancements to further improve its performance and usability. One key area of advancement is real-time processing—integrating edge computing capabilities can significantly reduce latency and enable faster inference. Expanding the training dataset with a broader range of tumor types and imaging conditions will enhance the model's generalizability and detection accuracy. Future iterations of the system could also incorporate cloud-based diagnostics and integration with hospital management platforms to streamline clinical workflows. Additionally, implementing automated tumor classification to distinguish between benign and malignant tumors will support more comprehensive diagnosis and treatment planning.

APPENDIX I

SOURCE CODE

```
import os
import uuid
from flask import Flask, render_template, request, jsonify, send_file
from PIL import Image
from ultralytics import YOLO
from reportlab.lib.pagesizes import A4
from reportlab.pdfgen import canvas

# 🚀 Flask App Setup
app = Flask(__name__, template_folder="templates", static_folder="static")

# ☐ Load YOLOv8 Brain Tumor Detection Model
MODEL_PATH = "D:\\Projects\\Brain_tumor\\yolov8_brain_tumor.pt"
if not os.path.exists(MODEL_PATH):
    raise FileNotFoundError(f"❌ YOLOv8 model not found at:
{MODEL_PATH}")

model = YOLO(MODEL_PATH)  # 🔍 Initialize YOLOv8

# 🗀 Directory for Saving Reports
REPORT_DIR = "reports"
os.makedirs(REPORT_DIR, exist_ok=True)
```

```
# 🏷️ Class Labels from the Model
class_labels = model.names

# 🌐 Home Page Route
@app.route("/")
def index():
    return render_template("index.html")

# ☐ Prediction Route
@app.route("/predict", methods=["POST"])
def predict():
    if "file" not in request.files:
        return jsonify({"error": "⚠️ No file uploaded"}), 400

    file = request.files["file"]
    if file.filename == "":
        return jsonify({"error": "⚠️ No selected file"}), 400

    try:
        # 💾 Save Uploaded Image
        image_id = str(uuid.uuid4())
        image_path = os.path.join("static", f"{image_id}.jpg")
        file.save(image_path)

        # 🔍 Run YOLOv8 Detection
        results = model(image_path)[0]

        if len(results.boxes) == 0:
            label = "No Tumor Detected"
```

```
            confidence = "N/A"
        else:
            top_result = results.boxes.conf.max()
            top_class = int(results.boxes.cls[results.boxes.conf.argmax()])
            label = class_labels[top_class]
            confidence = round(float(top_result) * 100, 2)

        # 📝 Generate PDF Report
        report_filename = f"{image_id}_report.pdf"
        report_path = os.path.join(REPORT_DIR, report_filename)
        generate_report(report_path, image_path, label, confidence)

        # ✅ Return Response
        return jsonify({
            "prediction": f"{label} Detected" if confidence != "N/A" else label,
            "confidence": confidence,
            "report_url": f"/download_report/{report_filename}"
        })

    except Exception as e:
        return jsonify({"error": f"💥 {str(e)}"}), 500

# ☐ PDF Report Generation Function
def generate_report(report_path, image_path, label, confidence):
    c = canvas.Canvas(report_path, pagesize=A4)
    c.setTitle("☐ Brain Tumor Detection Report")

    # ☐ Title Section
    c.setFont("Helvetica-Bold", 20)
```

```
    c.drawString(150, 800, "☐ Brain Tumor Detection Report")

    # 🆔 Patient Info
    c.setFont("Helvetica", 12)
    c.drawString(50, 750, f"Patient ID:
{os.path.basename(report_path).split('_')[0]}")
    c.drawString(50, 720, f"Tumor Prediction: {label}")
    c.drawString(50, 700, f"Confidence: {confidence}%")
    # 🖼️ MRI Scan
    c.drawString(50, 670, "Uploaded MRI Scan:")
    c.drawImage(image_path, 50, 400, width=300, height=250)

    # ⚠️ Disclaimer
    c.setFont("Helvetica-Oblique", 10)
    c.drawString(50, 100, "⚠️ This AI result should be verified by a medical professional.")
    c.save()

# 📥 Route to Download PDF Report
@app.route("/download_report/<filename>")
def download_report(filename):
    path = os.path.join(REPORT_DIR, filename)
    if os.path.exists(path):
        return send_file(path, as_attachment=True)
    return "❌ Report not found.", 404
# ☐💻 Run the Flask App
if __name__ == "__main__":
    app.run(debug=True)
```

APPENDIX II

SCREENSHOTS

9.1 WEB INTERFACE OF BRAIN TUMOR DETECTION

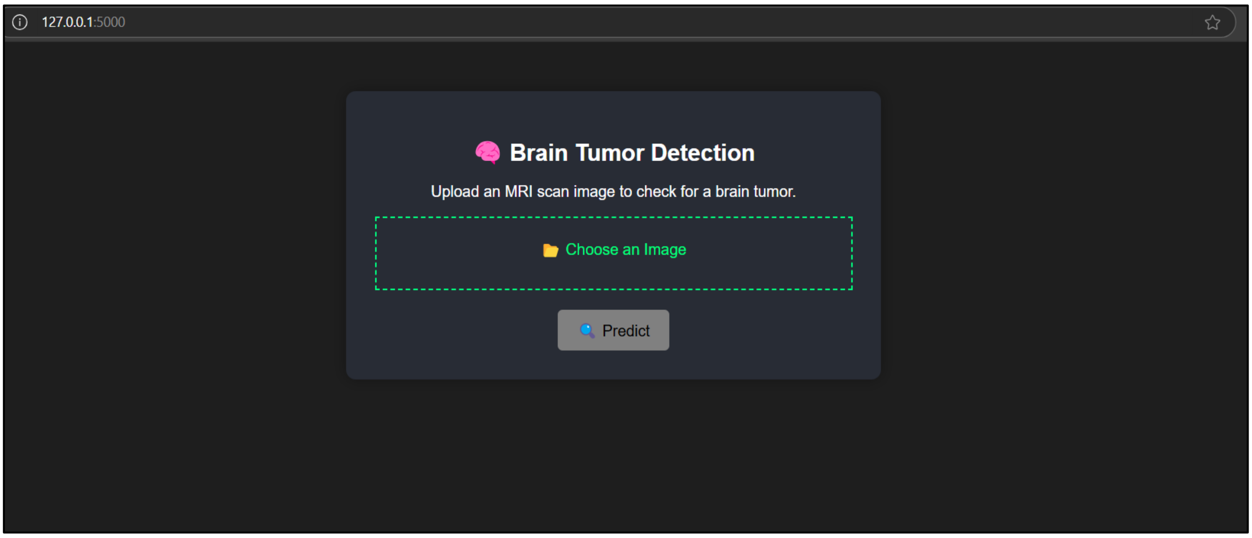

Fig.5 Image Uploading Page

This is the user interface of a brain tumor detection system where users can upload MRI scan images. The interface allows the user to select an image file and click the "Predict" button to analyze the scan for potential brain tumors.

9.2 RESULT OF DETECTION:

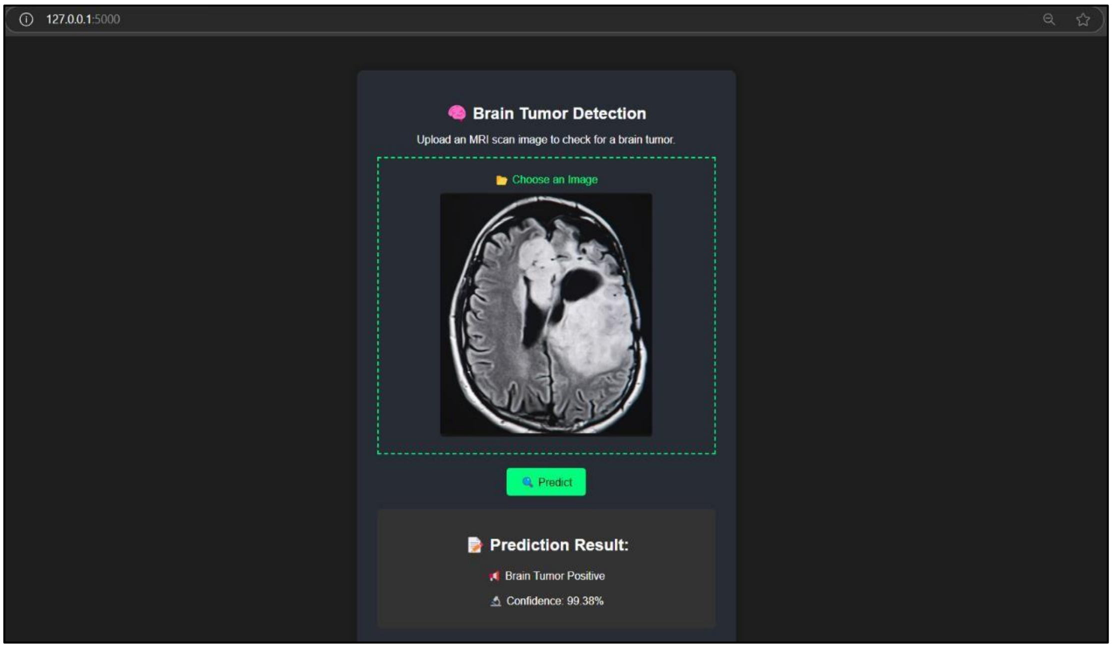

Fig.6 Positive Result of Detection

This interface displays the result of a brain tumor prediction. After uploading an MRI scan and clicking "Predict," the system analyzes the image and shows the result. In this case, it detected a brain tumor with a confidence level of 99.38%, indicating high accuracy in prediction.

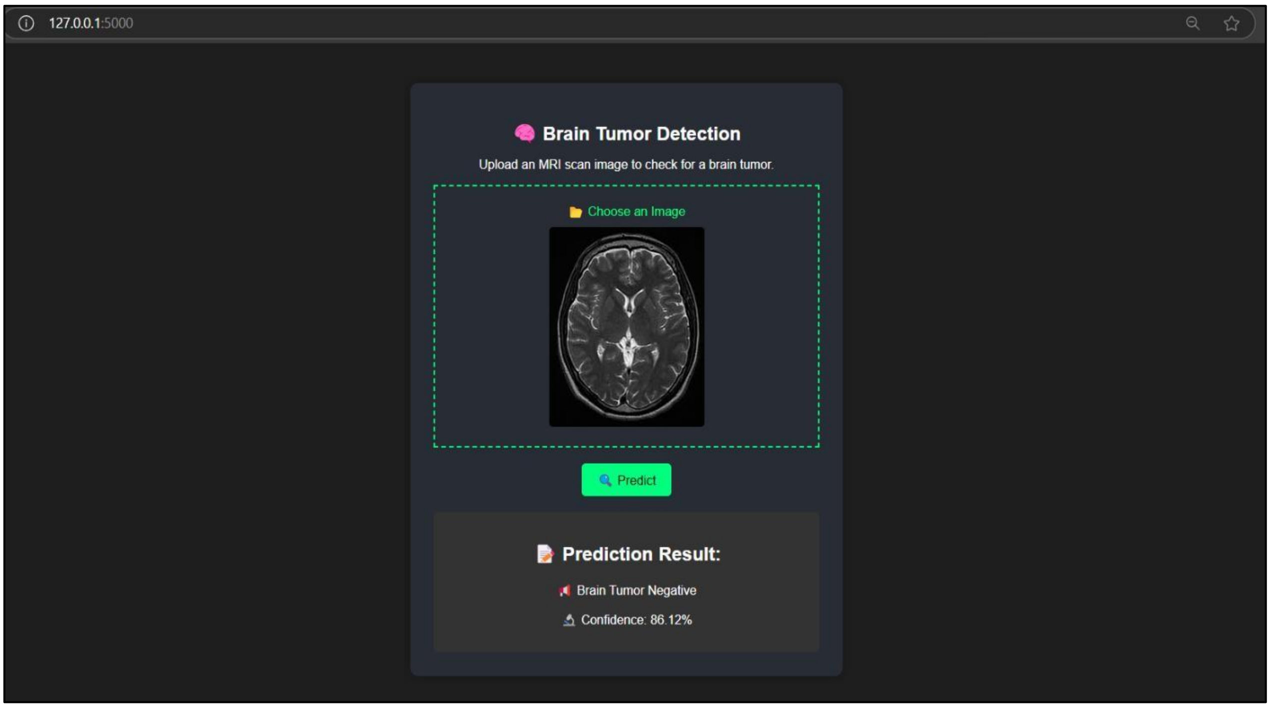

Fig.7 Negative Result of Detection

This interface displays the result of analyzing an MRI scan using the brain tumor detection model. After uploading the image and clicking "Predict," the system processes the scan and returns a negative result, indicating no tumor presence with a confidence level of 86.12%.

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